# INTRODUCTION TO DATA SCIENCE

## Opening Scenario

How many of you used Google Maps today?
How about Instagram, Netflix, or Swiggy?

All of you did, right?

Then here's some breaking news —
You already used Data Science today, without even knowing it.

**Think about it:**
- Google Maps knows the fastest route
- Netflix knows exactly what movie you'll love
- Swiggy knows when you'll feel hungry
- Instagram knows your mood

Now tell me, who told them all this?
Data told them.
And that's the magic of Data Science.

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## Weather Prediction Scenario

Imagine a weather forecasting system that predicts tomorrow's temperature, rainfall, and humidity.
How does it do that?
It collects data from satellites, weather sensors, and past

weather records, then applies statistical and machine learning models to predict what the weather will be tomorrow.
That's also Data Science in action.
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**Why Data Science? – Power of Data**

We are living in a data explosion era.

Every minute:
- 500K tweets are sent
- 1M Facebook posts are made
- 2 million Google searches happen
- 300 hours of YouTube videos are uploaded

Companies are literally drowning in data.
But who saves them?

**Data Scientists.**

They are like gold miners — digging valuable insights from useless dust.

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**Relatable Examples**

- Instagram's Explore page knows your crush (because you scroll too long there)
- Swiggy and Zomato know your cheat day (they push extra

burger ads on weekends)
- Amazon changes prices every 10 minutes using data
- Facebook can predict you're "in a relationship" even before you post it
- Netflix saves $1 billion every year just by recommending the right movies

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**When and Where Data Science is Used**

Industry: Entertainment
Use: Recommends movies
Example: Netflix knows you better than your mom

Industry: E-commerce
Use: Product suggestions
Example: Amazon says "You forgot toothpaste again!"

Industry: Healthcare
Use: Predicts diseases early
Example: FitBit warns before your doctor does

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**How Data Science Works – Simple Pipeline**

Data → Cleaning → Analysis → Machine Learning → Prediction → Decision

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**Mind-blowing Data Facts**

- 80% of a data scientist's time is spent cleaning data
- NASA uses Data Science to find planets
- Netflix saves $1 billion each year using Machine Learning models

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**What is Data Science?**

Definition:
Data Science means finding useful meaning from data.
It's the art of turning raw numbers into real-world decisions and predictions.

"We collect, clean, analyze, and predict — all using data."

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**Why Data Science?**

Because data is everywhere — but without analysis, it's just noise.
Companies need Data Science to make smarter, faster, and more accurate decisions.

"Without Data Science, businesses guess. With it, they know."

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**When Do We Use Data Science?**

Whenever we need to understand the past, act in the present, or predict the future using data.

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**Steps in Data Science**

Your goal is to make the best mango juice and predict which mango batch will give the sweetest taste.

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1. Data Extraction (Collecting Raw Mangos)

Meaning: Collecting raw data from multiple sources.

Juice Factory Analogy:
You get mangos from different farms, markets, and warehouses.
Some are small, some are overripe — all are raw materials for your juice.

Technologies Used:

- Databases: MySQL, MongoDB, PostgreSQL, Excel, CSV
- Big Data Tools: Apache Spark, Hadoop
- Web/Data Extraction: APIs, Web Scraping (BeautifulSoup, Requests, Selenium)
- Cloud Storage: Azure Data Lake, AWS S3
- Excel/CSV handling: Pandas, Power BI

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2. Data Cleaning (Removing Rotten Mangos)

Meaning: Fixing errors, removing missing or duplicate data.

Juice Factory Analogy:
You throw away rotten mangos, wash dirt, and remove insects. Similarly, you clean messy data — missing values, typos, duplicates.

Technologies Used:
- Python Libraries: Pandas, NumPy
- BI Tools: Power BI

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3. Feature Engineering (Extracting the Juice Pulp)

Meaning: Transforming raw data into useful features that help prediction.

Juice Factory Analogy:
You extract mango pulp, remove seeds, add sugar, or mix other fruits — improving the juice quality.
In data terms, you derive new columns or features like "average sweetness", "ripeness level", etc.

Technologies Used:
- Python: Pandas, NumPy, Scikit-learn

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4. Data Visualization (Tasting and Decorating the Juice)

Meaning: Understanding data patterns using visual graphs.

Juice Factory Analogy:
You taste the juice, adjust sugar, check color, and decorate the bottle — to see if it looks appealing.
Similarly, you visualize data to find insights like "which farm gives the sweetest mangoes?"

Technologies Used:
- Visualization Tools: Power BI, Tableau, Matplotlib, Seaborn, Plotly
- Dashboards: Power BI

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5. Machine Learning (Predicting the Best Mango Batch)

Meaning: Training algorithms to make predictions or classifications.

Juice Factory Analogy:
You analyze data of past mangoes — weight, color, sugar level — and train a model to predict which batch gives the sweetest juice next season.

Technologies Used:
- Python Libraries: Scikit-learn

Types:
- Regression: Predict sweetness level
- Classification: Good/Bad batch
- Clustering: Group similar mango farms

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6. Validation and Testing (Tasting by Customers)

Meaning: Checking how well your model performs on unseen data.

Juice Factory Analogy:
You give the juice to a few test customers to get feedback.

If they like it — your model (recipe) is ready. If not — you tune it.

Technologies Used:
- Evaluation Metrics: Accuracy

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7. Deployment (Selling the Juice in the Market)

Meaning: Making the model available for real-world use.

Juice Factory Analogy:
Now your mango juice bottle is ready and you ship it to stores so people can buy it.

Technologies Used:
- MLOps

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8. Maintenance (Checking Juice Quality Regularly)

Meaning: Monitoring model performance and updating with new data.

Juice Factory Analogy:
Over time, mango quality changes due to weather — you need to recheck recipes and improve flavor.
Similarly, models must be retrained with new data to stay

accurate.

Technologies Used:
- MLOps Tools

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**Comparison of AI, ML, DL, LLM, and Data Science**

Concept: AI (Artificial Intelligence)
What it Does: Thinks like humans
Example: Siri, Alexa

Concept: ML (Machine Learning)
What it Does: Learns from data
Example: Spam filter

Concept: DL (Deep Learning)
What it Does: Learns deep patterns
Example: Face recognition

Concept: LLM (Large Language Models)
What it Does: Understands language
Example: ChatGPT

Concept: Data Science
What it Does: Uses data to create insights using all of the above
Example: Netflix recommendations

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**Definitions and Examples**

1. Data Science
Definition: Data Science is the field that uses data to find insights and make predictions.
Example: Netflix uses Data Science to suggest movies you might like based on your past viewing habits.

2. Artificial Intelligence (AI)
Definition: AI is the science of making machines think and act like humans — learning, reasoning, and problem-solving.
Example: Voice assistants like Alexa or Siri understand your speech and respond — that's AI in action.

3. Machine Learning (ML)
Definition: ML is a subset of AI that allows systems to learn from data without being explicitly programmed.
It builds models that improve automatically with experience.
Example: Email spam filters learn which emails are spam by studying previous examples.

4. Deep Learning (DL)
Definition: DL is a subset of ML that uses neural networks (like the human brain) to handle complex data such as images, text, and sound.
Example: Face recognition on your phone uses Deep Learning to identify your face.

5. Large Language Models (LLM)

Definition: LLMs are a type of Deep Learning model trained on massive text data to understand and generate human-like language.
Example: ChatGPT can write, summarize, translate, and answer questions based on natural language.

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**Relationship Between AI, ML, and DL**

Your doubt:
"If Deep Learning (DL) is more advanced than Machine Learning (ML), then why do we call DL a subset of ML?"

Explanation:
Think of it like this:

AI → ML → DL

Just like:
Human → Student → Engineering Student

Every Engineering Student is a Student,
but not every Student is an Engineering Student.

Similarly:
Every DL model is an ML model,
but not every ML model is a DL model.

and ?????????

------------------------That's it-------------------------